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MILITARY SPECIFICATION

TRANSFORMER, DISCRIMINATOR

(For Torpedo Mark 37 Mods 0, 2, and 3)

This specification has been approved by the Naval Ordanance Systems Command, Department of the Navy

1. SCOPE

1.1 This specification establishes the requirements for the procurement of a Discriminator Transformer, NAVORD Dwg 1563729, which is used in the enabler assembled unit of an acoustic panel.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation forbids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

PPP-B-621	Boxes, Wood, Nailed and Lock-Corner

Military

MIL-E-1	Electronic Tubes; General Specification for
MIL-C-5	Capacitors, Fixed, Mica-Dielectric; General Specification for
MIL-C-25	Capacitors, Fixed, Paper-Dielectric, Direct Current (Hermetically Sealed in Metallic Cases)
MIL-P-116	Preservation, Methods of

FSC 1355

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MIL-R-10509	Resistors, Fixed Film (High stability), General Specification for
MIL-S-19500	Semiconductor Devices, General Specifications for
MIL-E-82590	Environmental Requirements and Tests (For Torpedo Mk 37 Components and Assemblies)

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-147	Palletized and Containerized Unit Loads, 40″ x 48″ Pallets, Skids, Runners, or Pallet-Type Base
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts

DRAWINGS

Naval Ordnance Systems Command

NAVORD dwg 1563729	Transformer, Discriminator

(Copies of specifications, standards, drawings, and publications required by the contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

National Bureau of Standards Publications

Handbook H28	Screw-Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing office, Washington, D.C. 20402.)

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United States of America Standards Institute

USAS Y32.2 1967 Graphic Symblos for Electrical and Electronics Diagrams

(Application for copies should be addressed to DOD Single Stock Point, U.S. Naval supply Depot, Philadelphia Pennsylvania 19120.)

3. REQUIREMENTS

3.1 Description. The discriminator transformer is a plug-in assembly, composed of various electronic components, in a hermetically sealed metal container. It is used in a circuit which converts frequency deviation to equivalent direct current (dc) output voltages.

3.2 General requirements

3.2.1 Preproduction sample. Unless otherwise specified in the contract or order, preproduction samples of the transformer shall be manufactured using the methods and procedures purposed for the production lot. These units shall be tested as specified in Section 4 herein and are for the purpose of determining, prior to starting production) that the contractor's production methods are capable of producing transformers that comply with the technical requirements of the contract. See 4.3.1.

3.2.2 Materials. Unless otherwise specified in the contract, requisition, or order, all materials and purchased parts entering into the final assembly of the transformer shall conform strictly to the detailed requirements indicated on the applicable drawings, specifications, and publications listed in NAVORD Dwg 1563729.

3.2.3 Surface treatment, construction, and dimensions. Surface treatment, construction, and dimensions for the transformer shall be in accordance with the applicable drawings, specifications, and publications listed in NAVORD Dwg 1563729.

3.2.4 Moisture, fungus, and corrosion. The transformer shall be rendered moisture, fungus, and corrosion resistant in accordance with detail requirements specified on the drawings and specifications listed in NAVORD Dwg 1563729.

3.2.5 Marking. Marking of assemblies, subassemblies, and parts shall be in accordance with applicable drawings listed in NAVORD Dwg 1563729.

3.2.6 Threads. Unless otherwise specified, all threads shall be in accordance with National Bureau of Standards Handbook H28. The class or fit for threads shall be as specified on the applicable drawings.

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3.2.7 Conflicting requirements. In the event of conflict between the documents referenced herein and the contents of this specification, the contents of this specification shall apply.

3.3 Symbols. Interpret schematic diagram symbols in accordance with USAS T32.2 1967.

3.4 Performance requirements and product characteristics. The transformer shall meet the following performance requirements and product characteristics:

3.4.1 DC output voltages

3.4.1.1 The frequency necessary to produce zero voltage at output #1 (see Figure 1) of the transformer shall be 7000 plus or minus 50 Hertz (HZ), when measured as specified in 4.5.4.2.

3.4.1.2 The frequency necessary to produce zero voltage at output #2 (see Figure 1) of the transformer shall be within plus or minus 10 Hz of the frequency measured in 3.4.1.1, when measured as specified in 4.5.4.2.

3.4.2 Polarity of the output voltages

3.4.2.1 The polarity of the voltage at output #1 shall be negative when the frequency is less than the frequency in 3.4.1.1, when measured as specified in 4.5.4.3.

3.4.2.2 The polarity of the voltage at output #1 shall be positive when the frequency is greater than the frequency in 3.4.1.1, when measured as specified in 4.5.4.3.

3.4.2.3 The polarity of the voltage at output #2 shall be positive when the frequency is less than the frequency in 3.4.1.2, when measured as specified in 4.5.4.3.

3.4.2.4 The polarity of the voltage at output #2 shall be negative when the frequency is greater than the frequency in 3.4.1.2, when measured as specified in 4.5.4.3.

3.4.3 Sensitivity. The sensitivities shall be as follows, when measured as specified in 4.5.4.4.

3.4.3.1 At the plus 10 vdc output point and the minus 10 vdc output point, the frequency shall be 94.0 plus or minus 7.0 Hz away from the zero vdc frequency point.

3.4.3.2 At the points 400 Hz above and 400 Hz below the zero dc output frequency, the magnitude of the output voltage shall be 48.0 plus or minus 8.0 volts.

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3.4.3.3 At the points 800 Hz above and 800 Hz below the zero dc output frequency, the magnitude of the output voltage shall be 70.4 plus or minus 6.4 volts.

3.4.3.4 The output voltage at 800 Hz below the zero dc output frequency on output #1 shall be within 6.5 volts of the output voltage at 800 Hz above the zero dc output frequency on output #2.

3.4.4 Dielectric withstanding voltage. The dielectric withstanding voltage shall be as follows, when tested as specified in 4.5.4.5.

3.4.4.1 The dielectric withstanding voltage between the input and the output terminals shall be 350 vdc.

3.4.4.2 The dielectric withstanding voltage between the input terminals and the case shall be 350 vdc.

3.4.4.3 The dielectric withstanding voltage between the output terminals and the case shall be 200 vdc.

3.4.5 Insulation resistance. The Insulation resistance of the transformer shall be 10,000 megohms minimum, when measured as specified in 4.5.4.6.

3.5 Environmental requirements. The transform shall meet the other applicable requirements of this specification during or after the tests defined or specified in 4.4.

3.5.1 Vibration. The transformer shall meet the following vibration requirements:

3.5.1.1 Vibration, nonoperating. The transformer shall show no damage and shall meet all the performance requirements and product characteristics of this specification after being tested as in 4.4.1.2.

3.5.1.2 Vibration, operating. The transformer shall show no damage and shall meet all the performance requirements and product characteristics of this specification during and after the test specified in 4.4.1.30

3.5.1.3 Vibration, transportation. The transformer shall show no damage and shall meet all the performance requirements and product characteristics of this specification after being tested as in 4.4.1.4.

3.5.2 Shock, transportation. The transformer shall show no damage and shall meet all the performance requirements and product characteristics of this specification after being tested as in 4.4.2.

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3.5.3 Temperature and humidity

3.5.3.1 High temperature and humidity, nonoperating. The transformer shall show no damage and shall meet all the performance requirements and product characteristics of this specification after being tested as in 4.4.3.1.

3.5.3.2 High temperature, operating. The transformer shall be capable of meeting all the performance requirements of this specification while undergoing the test specified in 4.4.3.2.

3.5.3.3 Low temperature, nonoperating. The transformer shall meet all the performance requirements and product characteristics of this specification after being tested as in 4.4.3.3.

3.5.3.4 Low temperature, operating. The transformer shall be capable of meeting all the performance requirements of this specification while undergoing the test specified in 4.4.3.4.

3.5.4 Immersion. The transformer shall show no evidence of leakage during the immersion test specified in 4.4.4 and thereafter shall meet the performance requirements and product characteristics of this specification.

3.5.5 Operating life. The discriminator transformer, when tested as specified in 4.4.5, shall have an operating life of at least 800 cycles of operation.

3.6 Cleanup. Prior to and after final assembly, all parts, components, and the assembly shall be thoroughly cleaned of loose, spattered, or excess solder, metal chips, and other foreign matter. Burrs and sharp edges as well as excessive rosin flash shall be removed.

3.7 Workmanship. The transformer, including all parts and accessories, shall be constructed and finished in a manner to ensure compliance with all requirements of this specification. Particular attention shall be paid to neatness and thoroughness of soldering, wiring, impregnation of coils, marking of parts and assemblies, plating, painting, riveting, machine screw assemblage, welding, brazing, and freedom of parts from burrs and sharp edges. The standards of workmanship exhibited in any aproved preproduction, subject to any qualification stated in the Government's notice of approval, shall be determinative of the requirements of the contract relative to workmanship insofar as not specifically covered by applicable specifications.

4. QUALITY ASSURANCE PROVISIONS

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4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. Inspection of the transformer shall be classified as follows:

(a) Preproduction inspection (paragraph 4.4)

(b) Quality conformance inspection (paragraph 4.5)

(c) Periodic production inspection (paragraph 4.4)

4.3 Sampling

4.3.1 Preproduction. A preproduction sample of five transformers, manufactured in accordance with 3.2.1, shall be subjected to the preproduction tests detailed in 4.4 at an activity designated by the procuring activity. Acceptance of the preproduction sample shall be based cm no defects in the sample. Further production of the transformers by the supplier, prior to approval of the preproduction sample, shall be at the supplier’s risk.

4.3.2 Lot size. Unless otherwise specified in the contract or order, each shipments of product shall be a separate "inspection lot" and may differ from the quantity specified in the total order or contract. In any event, all units of a single “inspection lot” shall be made to the same design, with the same materials and manufacturing techniques, and shall be part of one continuous production run.

4.3.3 Periodic production samples. From each succeeding lot or grand lot of 300 or less that have passed the quality conformance inspection of 4.5, a sample shall be selected in accordance with MIL-STD-105 Inspection Level S-2 for submission to an activity designated by the procuring activity to determine compliance of the sample with the requirements of the contract, specification, and drawings. The transformers shall be considered satisfactory if they meet all specification and pertinent drawings requirements and withstand the tests specified in 4.4 and 4.5. Failure of the periodic production sample to comply with these requirements will result in the rejection of the lot, as determined by the procuring activity.

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4.4 Reproduction and periodic production inspection. The preproduction and periodic production samples, after satisfactorily passing the quality conformance inspections detailed in 4.5, shall be subjected to the following tests and examinations. Certified inspection data covering the results of preproduction and periodic production inspections shall be forwarded to the procuring actitity.

4.4.1 Vibration

4.4.1.1 Mounting. The transformer, less pack or packaging, shall be mounted to a test fixture as shown in Figure 2 and tested as specified in 4.4.1.2 and 4.4.1.3.

4.4.1.2 Vibration, nonoperating. The transformer shall be subjected to the “Vibration, nonoperating” test of MIL-E-82590 and thereafter shall meet the requirements of 3.5.1.1.

4.4.1.3 Vibration, operating. The transformer shall be subjected to the “Vibration, operating” test of MIL-E-82590 and during and thereafter shall meet the requirements Of 3.5.1.2.

4.4.1.4 Vibration, transportation. The transformer shall be subjected to the “Vibration, transportation”, test of MIL-E-82590 and thereafter shall meet the requirements of 3.5.1.3.

4.4.2 Shock, transportation. The transformer shall be subjected to the "shock, transportation", test of MIL-E-82590 and thereafter shall meet the requirements of 3.5.2.

4.4.3 Temperature and humidity

4.4.3.1 High temperature and humidity, nonoperating The transformer shall be subjected to the “High temperature and humidity, nonoperating” test of MIL-E-82590 and thereafter shall meet the requirements of 3.5.3.1.

4.4.3.2 High temperature, operating. The transformer shall be subjected to the "High temperature, operating” test of MIL-E-82590 and, while at the test temperature, shall meet the requirements of 3.5.3.2.

4.4.3.3 Low temperature, nonoperating. The transformer shall be subjected to the “Low temperature, nonoperating” test of MIL-E-82590 and thereafter shall meet the requirements of 3.5.3.3.

4.4.3.4 Low temperature, operating. The transformer shall be subjected to the “Low temperature, operating" test of MIL-E-82590 and, while at the test temperature, shall meet the requirements of 3.5.3.4.

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4.4.4 Immersion. The transformers shall show no evidence of leakage while being subjected to the immersion test specified in MIL-STD-202, Method 104, Condition A, and thereafter shall meet the performance requirements and product characteristics of this specification.

4.4.5 Operating life. The test for operating life shall be conducted as follows:

4.4.5.1 Connect the transformer as shown in Figure 1. Apply heater voltage to the vacuum tube and allow it to warm up for at least 30 seconds. Set the signal generator to 7800 Hz plus or minus 20 Hz at an amplitude of 15 plus or minus 1 vrms. Apply 150 vdc as shown in Figure 1 and allow the circuit to operate for 30 plus or minus 1 minutes. Then remove the 150 vdc and allow the circuit to rest for at least 1 hour. This completes one cycle of operation.

4.4.5.2 Repeat 4.4.5.1, alternating the frequency of the signal. generator between 7800 plus or minus 20 Hz and 6200 plus or minus 20 Hz, until 800 cycles of operation are completed, after which the transformer shall meet the requirements of 3.4.1 through 3.4.5.

4.4.5.3 The vacuum tube and other test circuit components maybe replaced as necessary to allow the continuance of this test.

4.4.6 Waiver of tests. Tests specified in 4.4.1.4 and 4.4.2 above may be waived by the procuring agency for Level C packs only, provided the contractor guarantees safe delivery to the using activity.

4.4.7 Functional tests. At the conclusion of the tests specified in 4.4.1.4 and 4.4.2 above, functional and other tests, as applicable, shall be conducted on the transformer to determine freedom from operational malfunction and compliance with all performance requirements and product characteristics Of this specification.

4.5 Quality conformance inspection. Each transformer delivered under contract shall be subjected to the following tests and examinations. Failure to meet any requirement specified herein shall be considered cause for rejection.

4.5.1 Test conditions. Unless otherwise specified, the transformer shall be subjected to quality conformance tests under the following conditions:

4.5.1.1 Temperature. Room ambient 65 degrees F to 95 degrees F (18.3 degrees Centigrade (C) to 35 degrees C).

4.5.1.2 Altitude. Normal ground.

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4.5.1.3 Vibration. None.

4.5.1.4 Humidity. Room ambient to 95 percent relative maximum.

4.5.2 Test and inspection equipment and facilities. The manufacturer shall furnish and maintain all necessary test equipment and facilities and shall provide personnel for performing all quality conformance tests.

The test equipment shall be adequate in quantity and, when requirements are not specified, shall be Of sufficient accuracy and quality to permit performance of the required quality conformance tests.

4.5.3 Test equipment. The following items of test equipment are required to perform the quality conformance tests set forth in this specification:

4.5.3.1 Signal generator: Hewlett-Packard 205AH, or equivalent.

4.5.3.2 Electronic counter: Hewlett-Packard 522B, or equivalent.

4.5.3.3 Voltmeter: Ballantine Model 300, or equivalent.

4.5.3.4 Vacuum tube voltmeter: Hewlett-Packard Model 410B, or equivalent.

4.5.3.5 Insulation and dielectric tester.

4.5.3.6 Power source: 115 volts, 60 Hz, single phase.

4.5.3.7 Power supply: A power supply capable of supplying electrical power as specified in the following paragraphs. The power supply shall have switches in the output circuits capable of applying or removing power from the test circuit without securing the power supply. These switches shall be referred to as the "HEATER POWER" and "PLATE POWER" switches. Heater and plate power shall be considered applied when the switches are in the ON position and removed when in the OFF position. See Figure 1 for test circuit.

4.5.3.7.1 Heater power: The electrical power which is used to operate the 6AU6WB filament in the transformer test circuit shall meet the following requirements. All values specified shall be measured at the power supply terminals.

4.5.3.7.1.1 No-load voltage: With no load, the voltage shall not exceed minus 7.5 volts direct current (vdc).

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4.5.3.7.1.2 Current: The heater will require a current of 0.3 ampere.

4.5.3.7.1.3 Voltage: Under a load current of 0.3 ampere, the voltage shall be minus 6.25 plus or minus 0.10 vdc.

4.5.3.7.1.4 Internal impedance: The impedance which the power source presents to the filament at a load current of 0.3 ampere shall not exceed 400 ohms at 7 kiloHertz (kHz)

4.5.3 .7.1.5 Alternating current voltage components: Under a load current of 0.3 ampere, the alternating current (ac) voltage components shall not exceed the following:

30 Hz to 150 kHz	0.09 volt root mean square (vrms)
6.0 kHz to 8.0 kHz	20 microvolts rms

4.5.3.7.2 Plate power: The electrical power which is used to supply the 6AU6WB plate voltage in the transformer test circuit shall meet the following requirements. All values specified shall be measured at the power supply terminals

4.5.3.7.2.1 No load voltage: With no load, the voltage shall not exteed 165 vdc.

4.5.3.7.2.2 Current: The test circuit will require a current of 10 milliamperes (ma).

4.5.3 .7.2.3 Voltage: Under a load current of 10 ma, the voltage shall be 150 plus or minus 1 tic.

4.5.3.7.2.4 Internal impedance: The impedance which the power source presents to the test circuit at a load current of 10 ma shall not exceed 5 ohms at 7 kHz.

4.5.3 .7.2.5 Alternating current voltage components: Under a load current of 10 ma, the ac voltage components shall not exceed the following :

30 Hz to 150 kHz	0.2 vrms
6 kHz to 8 kHz	200 microvolts rms

4.5.3.8 Electron tube: MIL-E-1, JAN 6AU6WB.

4.5.3.9 Resistor: 150 ohms plus or minus 1 percent 1/2 W; MIL-R-10509 designation, RN65D1500F.

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4.5.3.10 Resistor: 220 ohms plus or minus 1 percent, 1/2 W; MIL-R-10509 designation, RN65D2200F.

4.5.3.11 Resistor: 1000 ohms plus or minus 1 percent 1/2 W; MIL-R-10509 designation, RN65D1001F.

4.5.3.12 Resistor: 100,000 ohms plus or minus 1 percent, 1/2 W; MIL-R-10509 designation, RN65D1003F.

4.5.3.13 Resistor: 200,000 ohms plus or minus 1 percent l12 W; MIL-R-10509 designation, RN65D2003F.

4.5.3.14 Resistor: 240,000 ohms plus or minus 1 percent, 1/2 W; MIL-R-10509 designation, RN65D2403F (6 required).

4.5.3.15 Resistor: 470,000 ohms plus or minus 1 percent, 1/2 W; MIL-R-10509 designation, RN65D4703F.

4.5.3.16 Resistor: 750,000 ohms plus or minus 0.1 percent, 1/2 W; MIL-R-10509 designation, RN65F7503B (4 required).

4.5.3.17 Resistor: 2.33 meg plus or minus 1 percent, 500 V; MIL-R-10509 designation, RN70DF2334F (two required).

4.5.3.18 Resistor: 750,000 ohms plus or minus 0.1 percent, 1/2 W; MIL-R-10509 designation, RN65F703B.

4.5.3.19 Capacitor: 0.22 uf plus or minus 10 percent; MIL-C-25 designation, CP09A1KC224K.

4.5.3.20 Capacitor: 470 uuf plus or minus 2 percent; MIL-C-25 designation, CP09A1KC471G (4 required).

4.5.3.21 Capacitor: 2700 uuf plus or minus 2 percent, MIL-C-5 designation, CM20B272G (required).

4.5.3.22 Capacitor: 1000 uuf plus or minus 10 percent, 500 V, MIL-C-25 designation, CP09A1KC102K.

4.5.3.23 Semiconductor device: MIL-S-19500 JAN 1N458.

4.5.3.24 Semiconductor device: A matched set of four 1N1257 diodes, with a forward current flow of 1 milliampere: the voltage drop across the diodes shall be within 1.0 millivolt of each other.

4.5.4 Test procedure. The following tests constitute the quality conformance tests for the transformer.

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4.5.4.1 Connect the following test equipment to the power source (4.5.3.6):

(a) Signal generator (4.5.3.1)

(b) Electronic counter (4.5.3.2)

(c) Vacuum tube voltmeter (4.5.3.4)

(d) voltmeter (4.5.3.3)

(e) Insulation tester (4.5.3.5)

(f) Power supply (4.503.7)

4.5.4.1.1 Set the power switches to “ON” and allow the test equipment to warm up long enough to stabilize.

4.5.4.2 To measure the dc output voltage as set forth in 3.4.1:

1.5.4.2.1 Connect the transformer as shown in Figure 1.

4.5.4.2.2 Adjust the signal generator (4.5.3.1) for 15 vrms output at a frequency of 7000 plus or minus 50 Hz.

4.5.4.2.3 set the power Supply (4.5.3.7) to plus 150 plus or minus 1 vdc.

4.5.4.2.4 Set the dc voltmeter (4.5.3.3) to a convenient voltage range and observe the dc voltage at output #1.

4.5.4.2.5 Adjust the frequency until the voltage is zero volts and record this frequency.

4.5.4.2.6 The frequency shall be within the limits specified in 3.4.1.1.

4.5.4.2.7 Connect the de voltmeter to output #2 and repeat the procedure as set forth in 4.5.4.2.

4.5.4.2.8 The frequency shall be within the limits specified in 3.4.1.2.

4.5.4.3 To determine the polarities of the output voltages as set forth in 3.4.2.

4.5.4.3.1 Connect the dc voltmeter between output #1 of the transformer and ground.

4.5.4.3.2 Vary the frequency of the signal generator from 6000 to 8000 Hz and observe the polarity of the output voltage.

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4.5.4.3.3 The polarity shall be as specified in 3.4.2.1 and 3.4.2.2.

4.5.4.3.4 Repeat the above procedure at output #2.

4.5.4.3.5 The polarity shall be as specified in 3.4.2.3 and 3.4.2.4.

4.5.4.4 To measure the sensitivities as set forth in 3.4.3:

4.5.4.4.1 Connect the transformer as shown in Figure 1.

4.5.4.4.2 Adjust the signal generator to provide 15 vrms at approximately 7000 Hz.

4.5.4.4.3 Connect the dc voltmeter to output #2 of the transformer.

4.5.4.4.4 Vary the oscillator frequency 1000 Hz above and below the center frequency for zero volt dc output. Read and record enough voltmeter readings over the frequency range to establish the shape of the frequency response curve.

4.5.4.4.5 At points 400 Hz above and 400 Hz below the zero volt dc output frequency, the magnitude of the output voltage shall be 48.0 plus or minus 8.0 volts.

4.5.4.4.6 At points 800 Hz above and 800 Hz below the zero dc output frequency, the magnitude of the output voltage shall be 70.4 plus or minus 6.4 volts.

4.5.4.4.7 At the plus 10 volts dc output point and the minus 10 volts dc output point, the frequency shall be 94 plus or minus 7 Hz above and below the zero vdc frequency.

4.5.4.4.8 Repeat the procedure of 4.5.4.4 with the dc voltmeter connected to output #1 of the transformer. The sensitivity shall be within the limits of output #2.

4.5.4.4.9 The output voltage at 800 Hz below the zero dc output frequency on output #1 shall be within 6.5 volts of the output voltage at 800 Hz above the zero dc output frequency on output #2.

4.5.4.5 To measure the dielectric withstanding voltage as set forth in 3.4.4.

4.5.4.5.1 Connect the insulation tester to pins 1 and 4 of the transformer.

4.5.4.5.2 Apply a potential of 350 vdc for a period of one minute. There shall be no evidence of breakdown.

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4.5.4.5.3 Connect the insulation tester to pin 1 and the case (mounting stud)) of the transformer.

4.5.4.5.4 Apply a potential of 350 vdc for a period of one minute. There shall be no evidence of breakdown.

4.5.4.5.5 Connect the insulation tester to pin 3 and the case (mounting stud) of the transformer.

4.5.4.5.6 Apply a potential of 200 vdc far a period of one minute. There shall be no evidence of breakdown.

4.5.4.6 To measure the isulation resistance as set forth in 3.4.5: (This test is to be performed after 4.5.4.5)

4.5.4.6.1 Connect the insulation tester (4.5.3.5) to pins 1 and 4 of the transformer.

4.5.4.6.1.1 Apply a potential of 100 vdc for two minutes.

4.5.4.6.1.2 Measure the resistance at a test potential of1 00 vdc after the two-minute period has elapsed.

4.5.4.6.1.3 The resistance shall meet the requirements of 3.4.5.

4.5.4.6.2 Connect the insulation tester to pin 1 and the case (mounting stud) of the transformer.

4.5.4 .6.2.1 Repeat the procedures of 4.5.4.6.1.1 and 4.5.4.6.1.2.

4.5.4.6.2.2 The resistance shall meet the requirements set forth in 3.4.5.

4.5.4.6.3 connect the insulation tester to pin 3 and the case (mounting stud) of the transformer.

4.5.4.6.3.1 Repeat the procedures of 4.5.4.6.1.1 and 4.5.4.6.1.2

4.5.4.6.3.2 The resistance shall meet the requirements set forth in 3.4.5.

4.5.4.7 Visual and dimensional examinations. Each transformer shall be examined to determine conformance with the requirements for workmanship and dimensions.

4.5.4.8 Packaging, packing, and marking. Examine the packaging, packing, and marking to determine conformance with the requirements of Section 5.

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5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. The levels of preservation and packaging shall be as specified (see 6.2).

5.1.1 Level A

5.1*1.1 Unit packaging. The transformer shall be packaged in accordance with Method IIB of MIL-P-116.

5.1.2 Level B. Not applicable.

5.1.3 Level C.

5.1.3.1 Unit packaging. The transformer shall be packaged in accordance with Method III of MIL-P-116.

5.2 Packing. Packing shall be Level A or Level C, as specified (see 6.2).

5.2.1 Level A

5.2.1.1 Consolidation container. The transformers, packaged as described in 5.1.1.1, shall be packed in a wooden box in accordance with PPP-B-621, class 2.

5.2.1.2 Palletizing. When specified, Level A packs shall be palletized in accordance with MIL-STD-147, Pattern No. 95.

5.2.2 Level B. Not applicable.

5.2.3 Level C. The transformers, packaged in accordance with 5.1.3.1, shall be packed to afford protection against damage during shipment from the supply source to the first receiving activity for immediate use. Containers shall comply with common carrier regulations applicable to the mode of transportation.

5.3 Levels A or C cushioning. The unit package shall incorporate sufficient cushioning material, bracing, or other adequate shock absorbing devices to ensure that the transformer will met all the requirements of Section 3 after being tested in accordance with paragraph 4.4.1.4 (Vibration) and 4.4.2 (Shock). At the time of packaging or packing, the moisture content of the cushioning material shall be no more than 5 percent.

5.4 Marking

5.4.1 special markings. None, unless otherwsie specified.

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5.4.2 Normal markings. In addition to the markings required by the contract or order, unit packages and shipping containers shall be marked in accordance with the requirements of MIL-STD-129, including serial numbers.

6. NOTES

6.1 Intended use. The transformer is intended for use in Torpedo Mk 37 Mods 0, 2, and 3.

6.2 Ordering data. Procurement documents should specify the following:

6.2.1 Procurement Requirements.

(a) Title, number, and date of this specification.

(b) Any exceptions to this specification.

(c) Required level of preservation and packaging (see 5.1).

(d) Required level of packing, including palletizing (see 5.2).

(e) Assigned activity for preproduction and periodic production inspections.

(f) That additional samples required because of failure of any unit(s) to pass the prescribed periodic production inspections shall be provided and inspected at the expense of the supplier.

(g) Guarantee clause (see 6.3)

6.2.2 Contract data requirements. Items of deliverable data required by this specification are cited in the following paragraph herein:

Paragraph	Data Requirement	Applicable DID*
4.4	Preproduction and periodic production inspection	-

DIDS (Data Item Descriptions/DD Forms 1664) for the above requirements will be documented in the applicable ADL (Authorized Data List). Such data will be delivered as identified on completed (numbered) DIDS when specified on DD Forms 1423 (Contract Data Requirements Lists) and incorporated into applicable contracts.

6.2.3 The attention of the contracting officer is invited to the options in 4.4.6.

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6.3 Guarantee. The contract or order shall include a requirement for a guarantee stating that any transformer which fails to met the requirements of this specification within one year after delivery because of a defect in material or workmanship or because of the vendor's failure to comply with the requirements of the applicable drawings and specifications shall be replaced by the vendor at no cost to the Government.

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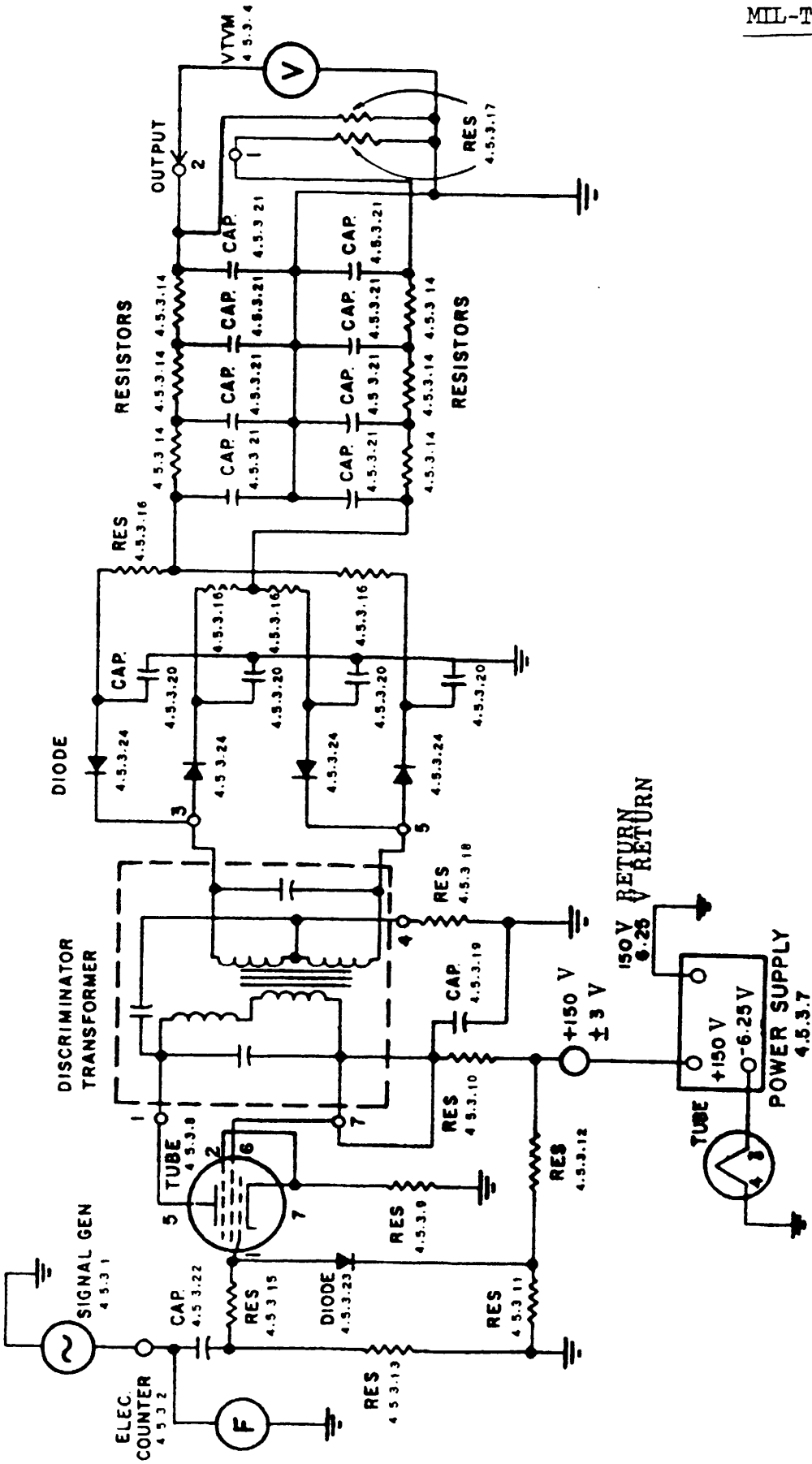
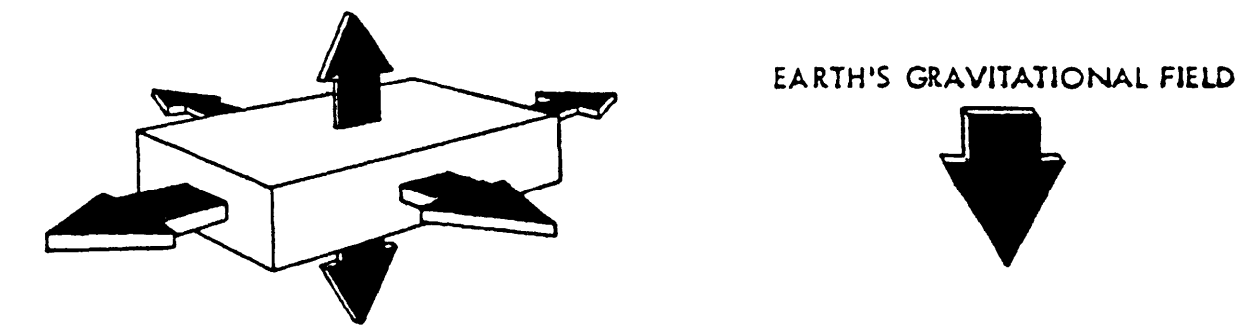
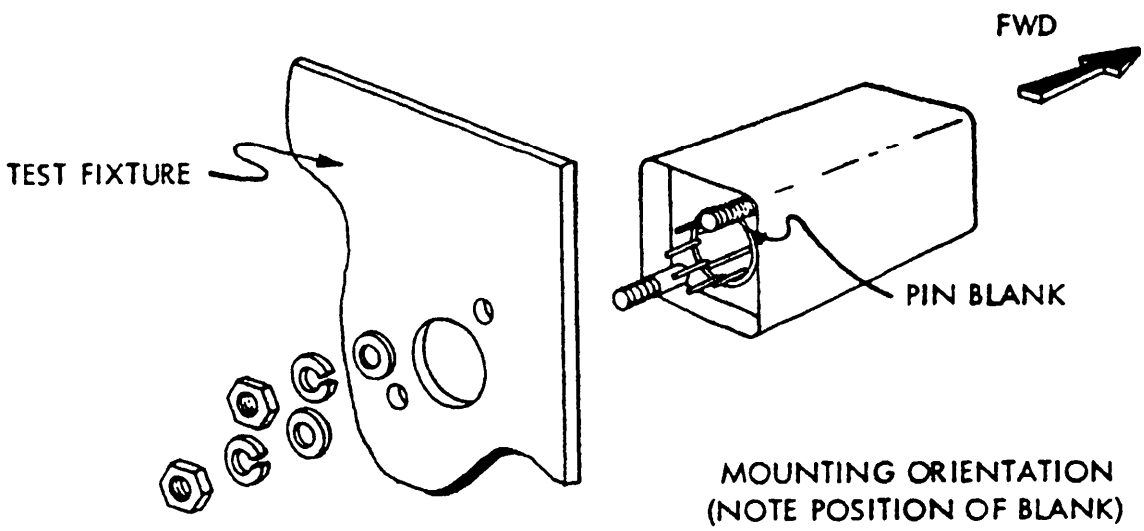


Figure 1. Test Circuit Schematic.

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FIGURE 2. DISCRIMINATOR TRANSFORMER
MOUNTING REQUIREMENTS





NOTES:

1. FOR VIBRATION, ORIENTATION OF TEST ITEM WITH RESPECT TO EARTH'S GRAVITATIONAL FIELD SHALL BE MAINTAINED FOR ALL TEST DIRECTIONS.

2. FOR VIBRATION, THE TEST ITEM SHALL BE SECURED TO THE TEST FIXTURE AS SHOWN.

3. THE TEST FIXTURE SHALL NOT ALTER THE INPUTS FROM VIBRATION EXCITER TO THE MOUNTING POINTS OF THE TEST ITEM.

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POSTAGE AND FEES PAID
NAVY DEPARTMENT

OFFICIAL BUSINESS

Commanding Officer
Naval Ordnance Station
Standization Division
ATTN: Code QA3
Indian Head, Maryland 20640

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE PLATE NO. 15419

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